STUDY GUIDE

Chapters 1–8

Combined compilation

CSMAXXXING

This compilation combines the source-based passages of Chapters 1–5 (from the supplied study guide, with verbatim excerpts from external sources) with the newly drafted Chapters 6–8. Every chapter opens with a mission and includes misconception checks, an activity with a feedback box, resources, and a source list. The missions, activities and glossary entries are drafted in text; the sourced passages are unchanged.

Table of Contents

1. Introduction to Artificial Intelligence 2
- 1.1 Defining AI and its Purpose 3
- 1.2 The Societal and Economic Impact of AI 3
- Activity 1A: Impact Map (paper activity) 3

2. The 5-Step Process: How AI Works 4
- 2.1 Input: Data Collection & Categorization 5
- 2.2 Processing: Pattern Recognition & Sorting 5
- 2.3 Outcomes: Predictions & Decision Making 5
- 2.4 Adjustments: Learning from Failures 5
- 2.5 Assessments: Synthesizing Insights & Feedback Loops 5
- Activity 2A: Sort the Fruit (paper activity) 6

3. Types of AI (Based on Capabilities) 7
- 3.1 Artificial Narrow AI (Weak AI) – Present-day Applications 8
- 3.2 Artificial General AI (Strong AI) – Theoretical Concepts 8
- 3.3 Super AI – Beyond Human Cognition 8
- Activity 3A: Knowledge Check (10 questions) 9

4. Practical Applications & Technologies 10
- 4.1 Computer Vision: Object Detection & Self-Driving Vehicles 11
- 4.2 Robotics: Industrial Automation & Healthcare Assistance 11
- 4.3 Expert Systems: Data Trends & Complex Problem Solving 11
- Activity 4A: Match the Technology (paper activity) 12

5. AI vs. Traditional Programming: Advantages & Disadvantages 13
- 5.1 Control and Transparency (The “Black Box” Problem) 14
- 5.2 Learning Capability & Unstructured Data Handling 14
- 5.3 Stability vs. Scalability 14
- 5.4 Efficiency, 24/7 Availability, and Workforce Implications 15
- Activity 5A: Choose Your Tool (paper activity) 15

Where Chapters 6–8 fit 17

6. Symbolic AI: Reasoning with Rules 18
- 6.1 What is Symbolic AI? 19
- 6.2 Strengths and limits 19
- Activity 6A: Build a Family Detective (SWISH / Prolog) 19

7. Connectionist AI: Learning from Data 21
- 7.1 What is connectionism? 22
- 7.2 How does it learn? 22
- 7.3 Symbolic vs connectionist 22
- 7.4 Types of neural networks (your mind map) 22
- Activity 7A: Be the Neuron (paper activity) 23
- Activity 7B: Playground Lab (TensorFlow Playground) 24

8. Behaviourist AI: Acting and Learning from Rewards 25
- 8.1 Behaviour-based AI 26
- Activity 8A: Layer the Vacuum (paper activity) 26
- 8.2 Reinforcement learning 27
- Activity 8B: Three Doors (dice game, exploration vs exploitation) 27

Supplementary Resources 28

Glossary 29

1. Introduction to Artificial Intelligence

Mission 1: The AI Spotter

Challenge: Social media feeds, customer-support chats and weather forecasts all use AI. Can you spot where AI is at work, and explain why one source compares it to electricity and the internet?

1.1 Defining AI and its Purpose

Artificial intelligence (AI) is a foundational technology that is supercharging other scientific fields and, like electricity and the internet, has the potential to transform societies, economies, and politics worldwide.

> Source: https://setr.stanford.edu/technology/artificial-intelligence/2026
> Authors: Co-chairs Condoleezza Rice, Jennifer Widom, and Amy Zegart. Director and Editor in Chief Herbert S. Lin. Managing Editor Martin Giles.

Key ideas:

- AI is a **foundational technology**: other scientific fields are being boosted by it.
- The source compares it to **electricity and the internet**.
- It has the potential to transform **societies, economies and politics** worldwide.

1.2 The Societal and Economic Impact of AI

The real-world potential of AI is immense. Applications of AI include diagnosing diseases, personalizing social media feeds, executing sophisticated data analyses for weather modeling and powering the chatbots that handle our customer support requests. AI-powered robots can even assemble cars and minimize radiation from wildfires.

> Chrystal R. China, 2024
> Source: https://www.ibm.com/think/insights/artificial-intelligence-advantages-disadvantages

Common misconception: “AI is just robots and chatbots.” Not true. The applications named above range from diagnosing diseases to weather modeling; chatbots and robots are only two of them.

Link to Chapter 5.4: automation can also have significant job loss implications for the workforce.

Activity 1A: Impact Map (paper activity)

Goal	See how widely AI is used, and practise saying who it could help and whom it could affect.
Preparation	Paper and a pen. About 10 minutes.
Task	See below.
Observation	Fill in the table.
Reflection	Which applications could change how people work? Use Section 5.4 for one worked example.
Feedback	Answer box.

Setup. Section 1.2 names six applications of AI. For each one, write the **field** it belongs to (choose from: health, social media, weather and climate, customer service, manufacturing, environment and safety), **one thing it could improve**, and whether it could **change jobs** (yes, no or maybe).

#	Application (from Section 1.2)	Field	One thing it could improve	Could it change jobs?
1	Diagnosing diseases	?	?	?
2	Personalizing social media feeds	?	?	?
3	Data analyses for weather modeling	?	?	?
4	Powering customer-support chatbots	?	?	?
5	AI-powered robots assembling cars	?	?	?
6	Minimizing radiation from wildfires	?	?	?

Feedback box

- Fields: 1 = health, 2 = social media, 3 = weather and climate, 4 = customer service, 5 = manufacturing, 6 = environment and safety.
- Improvements and job impact will vary, so this part is for discussion. Section 5.4 gives one worked example: companies using digital assistants to triage employee reports instead of delegating them to a human resources department. Customer-support chatbots (row 4) are the closest match in this list, because they also take over a task that people used to handle.
- Lesson: AI appears in many unrelated fields, which is why the source calls it a foundational technology.
- Limit: the table only uses the six applications named in Section 1.2; it is not a complete list of what AI does.

Quick check: Why does the source compare AI to electricity and the internet? (Hint: look at Section 1.1.)

Resources

- **Stanford Emerging Technology Review, Artificial Intelligence** (the source for Section 1.1): https://setr.stanford.edu/technology/artificial-intelligence/2026. It is a long report, so start with the AI section only.
- **What Is Artificial Intelligence? Crash Course AI #1** (Jabril Ashe, CrashCourse with PBS Digital Studios, 2019, about 11 minutes): https://thecrashcourse.com/courses/what-is-artificial-intelligence-crash-course-ai-1/. Watch the stretch from about 01:59 to 04:23, which a transcript summary lists as examples of AI in daily life. Viewing question: which of those examples are also in Section 1.2, and which are new? (Timestamps are from the YouTube version.)

Sources (Chapter 1)

- Stanford Emerging Technology Review, Artificial Intelligence (2026). https://setr.stanford.edu/technology/artificial-intelligence/2026
- Chrystal R. China (2024). https://www.ibm.com/think/insights/artificial-intelligence-advantages-disadvantages

2. The 5-Step Process: How AI Works

Mission 2: The Pass-or-Fail Machine

Challenge: A machine has to decide “pass” or “fail” for every item it sees, and it gets some wrong. Can you walk it through the five steps, then help it fix its mistakes?

Below, we’ll go through five steps that illustrate how AI works: inputs, processing, outcomes, adjustments, and assessments.

2.1 Input: Data Collection & Categorization

Data is first collected from various sources in the form of text, audio, video, and more. It is sorted into categories, such as those that can be read by the algorithms and those that cannot. You would then create the protocol and criteria for which data will be processed and used for specific outcomes.

2.2 Processing: Pattern Recognition & Sorting

Once data is gathered and input, the next step is to allow AI to decide what to do with the data. The AI sorts and deciphers the data using patterns it has been programmed to learn until it recognizes similar patterns in the data that is being filtered into the system.

2.3 Outcomes: Predictions & Decision Making

After the processing step, the AI can use those complex patterns to predict outcomes in customer behavior and market trends. In this step, the AI is programmed to decide whether specific data is a “pass” or “fail”—in other words, does it match previous patterns? That determines outcomes that can be used to make decisions.

2.4 Adjustments: Learning from Failures

When data sets are considered a “fail,” AI learns from that mistake, and the process is repeated again under different conditions. It may be that the algorithm’s rules must be adjusted to suit the data set in question or that the algorithm needs slight alteration. In this step, you might return to the outcomes step to better align with the current data set’s conditions.

2.5 Assessments: Synthesizing Insights & Feedback Loops

The final step for AI to complete an assigned task is assessment. Here, the AI technology synthesizes insights gained from the data set to make predictions based on the outcomes and adjustments. Feedback generated from the adjustments can be incorporated into the algorithm before moving forward.

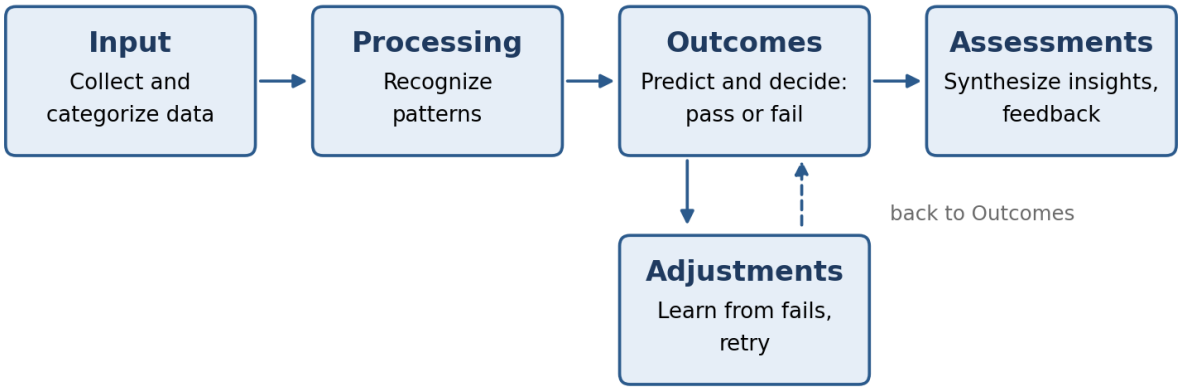


Figure 1. The 5-step process of how AI works

Coursera Staff, 2026
Source: https://www.coursera.org/articles/how-does-ai-work

Common misconception: "AI gets everything right the first time." Not true. The process has an Adjustments step because some data is a "fail" and the algorithm's rules may need to change (Section 2.4).

Link to Chapter 7.2: in a neural network, adjusting means changing the weights (you try this by hand in Activity 7A).

Activity 2A: Sort the Fruit (paper activity)

Goal	Walk one tiny example through all five steps, including a "fail" and an adjustment.
Preparation	Paper and a pen. About 15 minutes.
Task	See below.
Observation	Write your rule, the pass or fail result for each item, and the number correct in each round.
Reflection	Which step caught the mistakes? Did your adjustment create any new mistake?
Feedback	Answer box.

Setup. A toy "algorithm" must decide whether each item is an **apple**. It can only read two columns: **colour** and **shape**. It cannot read the smell.

Known examples (what the algorithm has seen before):

Example	Colour	Shape	Apple?
K1	red	round	yes
K2	red	round	yes
K3	yellow	long	no

Step 1, Input. Look at the table of new items below. Cross out any column the toy algorithm cannot read.

Step 2, Processing. From K1 to K3, write the pattern for "apple" as one rule.

Step 3, Outcomes. Apply your rule to N1 to N6. Write **pass** (matches the pattern) or **fail**, then check against the "Really an apple?" column.

Item	Colour	Shape	Smell	Really an apple?	Round 1: pass or fail	Correct?	Round 2: pass or fail	Correct?
N1	red	round	sweet	yes	?	?	?	?
N2	yellow	long	sweet	no	?	?	?	?
N3	green	round	sweet	yes	?	?	?	?
N4	red	long	spicy	no	?	?	?	?
N5	green	round	sweet	yes	?	?	?	?
N6	orange	round	citrus	no	?	?	?	?

Step 4, Adjustments. Look at the wrong results. Change your rule to fix them, then fill in the Round 2 columns.

Step 5, Assessments. Count the correct answers in each round. Did the adjustment cause a new mistake? If it did, go back to Outcomes and adjust again, as Section 2.4 describes.

Feedback box

- Step 1: the smell column cannot be read, so cross it out (Section 2.1 says data is sorted into what algorithms can read and what they cannot).
- Step 2: “red and round means apple”.
- Round 1 (rule: red and round): N1 pass (correct), N2 fail (correct), N3 fail (**wrong**, a green apple), N4 fail (correct), N5 fail (**wrong**), N6 fail (correct). Score: 4 out of 6.
- Step 4: drop the colour requirement, so the rule becomes “round means apple”. Round 2: N1 pass (correct), N2 fail (correct), N3 pass (correct), N4 fail (correct), N5 pass (correct), N6 pass (**wrong**, an orange). Score: 5 out of 6.
- Step 5: the score improved, but the adjustment created a new mistake, so the loop goes round again. A rule such as “round, and red or green” gets all six right. That only fits these six items; new items could break it again, which is why assessment keeps feeding back into adjustments.
- Limit: real AI learns patterns from large amounts of data (Section 5.2) rather than from a rule you write by hand. This is a simplified model of the five steps.

Quick check: Which two steps form a loop, and why? (Hint: look back at Section 2.4.)

Resources

- **Coursera, “How does AI work?”** (the source for this chapter): https://www.coursera.org/articles/how-does-ai-work. Use it to read the five steps in the original wording.
- **What Is Artificial Intelligence? Crash Course AI #1** (see Chapter 1 resources). Start at about 06:08, where the video says an AI does not really “know” anything when it is first created and has to learn from data. Viewing question: which of the five steps does that learning belong to?

Sources (Chapter 2)

- Coursera Staff, “How does AI work?” (2026). https://www.coursera.org/articles/how-does-ai-work

3. Types of AI (Based on Capabilities)

Mission 3: Name That AI

Challenge: A friend says, “I use an AI that can do anything a human can.” Which of the three types of AI is that, and does it exist today?

3.1 Artificial Narrow AI (Weak AI) – Present-day Applications

Artificial Narrow Intelligence, also known as Weak AI (what we refer to as Narrow AI), is the only type of AI that exists today. Any other form of AI is theoretical. It can be trained to perform a single or narrow task, often far faster and better than a human mind can.

However, it can’t perform outside of its defined task. Instead, it targets a single subset of cognitive abilities and advances in that spectrum. Siri, Amazon’s Alexa and IBM Watson® are examples of Narrow AI. Even OpenAI’s ChatGPT is considered a form of Narrow AI because it’s limited to the single task of text-based chat.

IBM Data and AI Team, 2024
Source: https://www.ibm.com/think/topics/artificial-intelligence-types

3.2 Artificial General AI (Strong AI) – Theoretical Concepts

Artificial General Intelligence (AGI), also known as Strong AI, is today nothing more than a theoretical concept. AGI can use previous learnings and skills to accomplish new tasks in a different context without the need for human beings to train the underlying models. This ability allows AGI to learn and perform any intellectual task that a human being can.

IBM Data and AI Team, 2024
Source: https://www.ibm.com/think/topics/artificial-intelligence-types

3.3 Super AI – Beyond Human Cognition

Super AI is commonly referred to as artificial superintelligence and, like AGI, is strictly theoretical. If ever realized, Super AI would think, reason, learn, make judgements and possess cognitive abilities that surpass those of human beings.

The applications possessing Super AI capabilities will have evolved beyond the point of understanding human sentiments and experiences to feel emotions, have needs and possess beliefs and desires of their own.

IBM Data and AI Team, 2024
Source: https://www.ibm.com/think/topics/artificial-intelligence-types

Level	Capability	Status
Super AI	Surpasses humans	Theoretical
General AI (AGI)	Any human intellectual task	Theoretical
Narrow AI	One task: Siri, Alexa, ChatGPT	Exists today

Figure 2. The three kinds of AI based on capabilities

IBM Data and AI Team, 2024
Source: https://www.ibm.com/think/topics/artificial-intelligence-types

Common misconception: “Siri and ChatGPT seem so smart that they must be general AI.” Not true. The source classes Siri, Alexa, IBM Watson and ChatGPT as Narrow AI, because each is limited to a defined task.

Link to Chapter 4: the practical applications there (computer vision, robotics, expert systems) all use Narrow AI.

Activity 3A: Knowledge Check (10 questions)

Goal	Check that you can tell the three types of AI apart and apply the definitions to examples.
Preparation	Paper and a pen. Try it without looking at the sections first.
Task	Answer all 10 questions below.
Observation	Write your answers, then mark them with the feedback box.
Reflection	Which questions confused you? Which definition do you need to reread?
Feedback	Answer box.

Instructions: Answer all 10 questions. Choose one answer for each multiple-choice question. For short-answer questions, answer in one or two sentences.

1. Multiple choice: Which type of AI is described as the only type that exists today in the study guide?
A. Super AI
B. Artificial General Intelligence (AGI)
C. Artificial Narrow AI
D. Artificial Superintelligence

2. Multiple choice: Which statement best describes Artificial Narrow AI (Weak AI)?
A. It can perform any intellectual task a human can.
B. It is designed or trained for a single or narrow task.
C. It always surpasses human cognition.
D. It has its own beliefs and desires.

3. Identify the type: A virtual assistant such as Siri or Amazon’s Alexa is given a defined task and cannot operate beyond that task. Which type of AI is this?
A. Super AI
B. Artificial General Intelligence
C. Artificial Narrow AI
D. None of the above

4. Multiple choice: According to the study guide, why is ChatGPT classified as Narrow AI?
A. It can perform every human intellectual task.
B. It is considered limited to text-based chat.
C. It surpasses human cognition in all areas.
D. It possesses independent desires.

5. Short answer: What is Artificial General Intelligence (AGI), and what key ability would it have?
Answer: _____

6. True or false: Artificial General Intelligence is described in the study guide as a technology that already exists.
Answer: _____

7. Scenario: An AI system can use what it has learned to perform new intellectual tasks in different contexts without humans training the underlying models. Which type of AI does this describe?
Answer: _____

8. Multiple choice: Which description best matches Super AI?
A. It performs only one narrow task.
B. It can perform any intellectual task that a human can.
C. Its cognitive abilities would surpass those of human beings.
D. It is the only type of AI that exists today.

9. Compare: State one difference between Artificial Narrow AI and Artificial General Intelligence (AGI).
Answer: _____

10. True or false: Super AI is theoretical and, if realized, would have cognitive abilities that surpass human beings.
Answer: _____

Feedback box

1. **C** — Artificial Narrow AI.
2. **B** — It can be trained to perform a single or narrow task and cannot perform outside its defined task.
3. **C** — Artificial Narrow AI.
4. **B** — The study guide describes ChatGPT as limited to the single task of text-based chat.
5. AGI is theoretical and could use previous learnings and skills to accomplish new tasks in different contexts, and perform any intellectual task that a human can.
6. **False** — AGI is described as a theoretical concept.
7. Artificial General Intelligence (AGI).
8. **C** — Its cognitive abilities would surpass those of human beings.
9. Narrow AI is limited to a single or narrow task; AGI would be able to learn and perform any intellectual task that a human can, including in new contexts.
10. **True.**

Resources

- **IBM, "What is artificial general intelligence (AGI)?"**: https://www.ibm.com/think/topics/artificial-general-intelligence. Optional further reading on Section 3.2; it is not used as a source in this guide.
- IBM Technology, The 7 Types of AI - And Why We Talk (Mostly) About 3 of Them

Sources (Chapter 3)

- IBM Data and AI Team (2024). https://www.ibm.com/think/topics/artificial-intelligence-types

4. Practical Applications & Technologies

Mission 4: The Smart Warehouse

Challenge: A warehouse wants machines that can see what is around them, move and sort goods, and spot trends in its data. Which kind of Narrow AI would do each job?

4.1 Computer Vision: Object Detection & Self-Driving Vehicles

Narrow AI applications with computer vision can be trained to interpret and analyze the visual world. This allows intelligent machines to identify and classify objects within images and video footage.

Applications of computer vision include:

- Image recognition and classification
- Object detection
- Object tracking
- Facial recognition
- Content-based image retrieval

Computer vision is critical for use cases that involve AI machines interacting and traversing the physical world around them. Examples include self-driving cars and machines navigating warehouses and other environments.

4.2 Robotics: Industrial Automation & Healthcare Assistance

Robots in industrial settings can use Narrow AI to perform routine, repetitive tasks that involve materials handling, assembly and quality inspections. In healthcare, robots equipped with Narrow AI can assist surgeons in monitoring vitals and detecting potential issues during procedures.

Agricultural machines can engage in autonomous pruning, moving, thinning, seeding and spraying. And smart home devices such as the iRobot Roomba can navigate a home's interior using computer vision and use data stored in memory to understand its progress.

4.3 Expert Systems: Data Trends & Complex Problem Solving

Expert systems equipped with Narrow AI capabilities can be trained on a corpus to emulate the human decision-making process and apply expertise to solve complex problems. These systems can evaluate vast amounts of data to uncover trends and patterns to make decisions. They can also help businesses predict future events and understand why past events occurred.

IBM Data and AI Team, 2024
Source: https://www.ibm.com/think/topics/artificial-intelligence-types

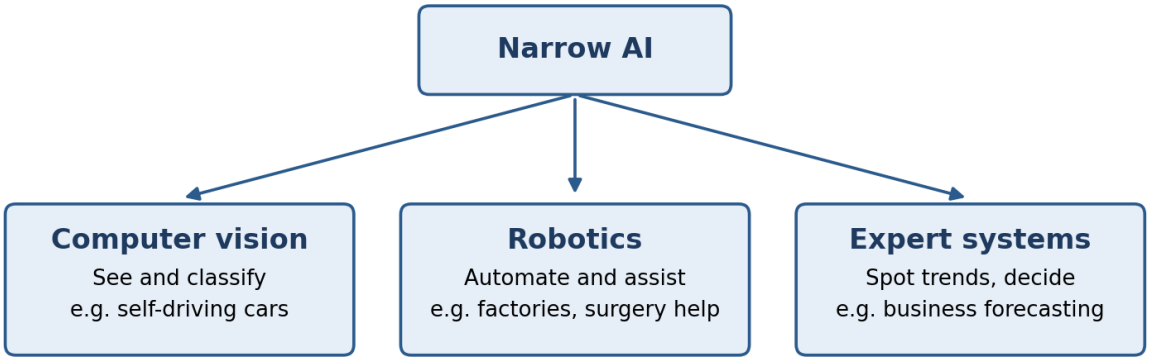


Figure 3. Practical applications of Narrow AI

IBM Data and AI Team, 2024
Source: https://www.ibm.com/think/topics/artificial-intelligence-types

Common misconception: "A robot that does many jobs must be general AI." Not true. The robots described here use Narrow AI for routine, defined tasks, and Section 3.1 says Narrow AI cannot perform outside its defined task.

Link to Chapter 3.1: every application in this chapter is Narrow AI, the only type that exists today.

Activity 4A: Match the Technology (paper activity)

Goal	Match real situations to the right kind of Narrow AI technology.
Preparation	Paper and a pen. About 10 minutes.
Task	See below.
Observation	Fill in the table.
Reflection	Which situation fits more than one technology? Why are all of them still Narrow AI?
Feedback	Answer box.

Setup. For each situation, write the technology it uses: **computer vision**, **robotics**, **expert system**, or **more than one**.

#	Situation	Technology?
1	A self-driving car identifies and classifies objects in video footage.	?
2	A machine navigates a warehouse and the environment around it.	?
3	A robot assists a surgeon by monitoring vitals and detecting potential issues.	?
4	An industrial robot performs quality inspections.	?
5	A farm machine autonomously seeds and sprays.	?
6	A system evaluates vast amounts of data to uncover trends and predict future events.	?
7	A Roomba navigates a home's interior using computer vision and data stored in memory.	?

Feedback box

- 1 = computer vision (Section 4.1).
- 2 = computer vision (Section 4.1 names machines navigating warehouses).
- 3 = robotics (Section 4.2, healthcare).
- 4 = robotics (Section 4.2, industrial settings).
- 5 = robotics (Section 4.2, agricultural machines).
- 6 = expert system (Section 4.3).
- 7 = **more than one**: it is a smart home robot (Section 4.2) that uses computer vision to navigate.
- Reflection answer: each situation is a defined task, so each is Narrow AI (Section 3.1). Technologies can also combine, as the Roomba shows.

Quick check: Which technology would you choose to forecast next month's stock needs, and which section says so? (Hint: look at Section 4.3.)

Resources

- **Robotics: Crash Course AI #11** (Jabril Ashe, CrashCourse with PBS Digital Studios, 2019, 9 min 52 s): https://www.pbs.org/video/robotics-11-ouvqgj/. The episode covers three robotics problems: localization, planning and manipulation. Viewing question: the series describes robots as built for specific tasks, such as vacuuming homes or assembling cars. Which type of AI from Chapter 3 does that match, and why? Watch the whole episode.

Sources (Chapter 4)

- IBM Data and AI Team (2024). https://www.ibm.com/think/topics/artificial-intelligence-types

5. AI vs. Traditional Programming: Advantages & Disadvantages

Mission 5: The Tool Picker

Challenge: A company has two projects: a rule checker where every decision must be traceable, and a tagger for thousands of photos and videos. Should each one use traditional programming or AI?

As with any technology, there are advantages and disadvantages of AI, when compared to traditional programming technologies. Aside from foundational differences in how they function, AI and traditional programming also differ significantly in terms of programmer control, data handling, scalability and availability.

Traditional	AI
Control: Full, traceable logic	Control: Black box, hard to explain
Data: Structured only	Data: Images, video, text
Learning: Manual updates	Learning: Learns from new data
Stability: Same result every time	Stability: May be inconsistent
Scalability: Needs more code	Scalability: Scales quickly
24/7 use: Often needs staff	24/7 use: Runs unattended

Figure 4. AI vs. traditional programming at a glance

5.1 Control and Transparency (The “Black Box” Problem)

Traditional programming offers developers full control over the logic and behavior of software, allowing for precise customization and predictable, consistent outcomes. And if a program doesn’t behave as expected, developers can trace back through the codebase to identify and correct the issue.

AI systems, particularly complex models like deep neural networks, can be hard to control and interpret. They often work like “black boxes,” where the input and output are known, but the process the model uses to get from one to the other is unclear. This lack of transparency can be problematic in industries that prioritize process and decision-making explainability (like healthcare and finance).

5.2 Learning Capability & Unstructured Data Handling

Traditional programming is rigid; it relies on structured data to execute programs and typically struggles to process unstructured data. In order to “teach” a program new information, the programmer must manually add new data or adjust processes. Traditionally coded programs also struggle with independent iteration. In other words, they may not be able to accommodate unforeseen scenarios without explicit programming for those cases.

Because AI systems learn from vast amounts of data, they’re better suited for processing unstructured data like images, videos and natural language text. AI systems can also learn continually from new data and experiences (as in machine learning), allowing them to improve their performance over time and making them especially useful in dynamic environments where the best possible solution can evolve over time.

5.3 Stability vs. Scalability

Traditional programming is stable. Once a program is written and debugged, it will perform operations the exact same way, every single time. However, the stability of rules-based programs comes at the expense of scalability. Because traditional programs can only learn through explicit programming interventions, they require programmers to write code at scale in order to scale up operations. This process can prove unmanageable, if not impossible, for many organizations.

AI programs offer more scalability than traditional programs but with less stability. The automation and continuous learning features of AI-based programs enable developers to scale processes quickly and with relative ease, representing one of the key advantages of AI. However, the improvisational nature of AI systems means that programs may not always provide consistent, appropriate responses.

5.4 Efficiency, 24/7 Availability, and Workforce Implications

Rules-based computer programs can provide 24/7 availability, but sometimes only if they have human workers to operate them around the clock.

AI technologies can run 24/7 without human intervention so that business operations can run continuously. Another of the benefits of artificial intelligence is that AI systems can automate boring or repetitive jobs (like data entry), freeing up employees' bandwidth for higher-value work tasks and lowering the company's payroll costs.

It's worth mentioning, however, that automation can have significant job loss implications for the workforce. For instance, some companies have transitioned to using digital assistants to triage employee reports, instead of delegating such tasks to a human resources department. Organizations will need to find ways to incorporate their existing workforce into new workflows enabled by productivity gains from the incorporation of AI into operations.

> Chrystal R. China, 2024
> Source: https://www.ibm.com/think/insights/artificial-intelligence-advantages-disadvantages

Common misconception: "AI is always better than traditional programming." Not true. Traditional programming gives full control and the same result every time; AI scales quickly and handles images, video and text. Each wins on different rows of Figure 4.

Link to Chapter 7.3: the symbolic vs connectionist table shows the same explainability trade-off.

Activity 5A: Choose Your Tool (paper activity)

Goal	Use the rows of Figure 4 to decide when traditional programming or AI fits a requirement.
Preparation	Paper and a pen. About 10 minutes.
Task	See below.
Observation	Fill in the table.
Reflection	Which requirement was hardest to decide? Which row of Figure 4 settled it?
Feedback	Answer box.

Setup. For each requirement, write **traditional** or **AI**, and name the row of Figure 4 (control, data, learning, stability, scalability or 24/7 use) that decided it.

#	Requirement	Traditional or AI?	Deciding row
1	Every decision must be traceable back through the logic (as in healthcare or finance).	?	?
2	Sort a large collection of photos and videos by what they show.	?	?
3	The program must give the exact same result every single time.	?	?
4	Operations must scale up quickly without writing code at scale.	?	?
5	The system must run around the clock without staff operating it.	?	?
6	The program must improve as new data arrives and the best solution changes.	?	?

Feedback box

- 1 = traditional (control: full, traceable logic; Section 5.1 says AI's black-box nature is a problem where explainability matters).
- 2 = AI (data: images, video, text; Section 5.2).
- 3 = traditional (stability: same result every time; Section 5.3).
- 4 = AI (scalability: scales quickly; Section 5.3).
- 5 = AI (24/7 use: runs unattended; Section 5.4).
- 6 = AI (learning: learns from new data; Section 5.2).
- Reflection: if a requirement matched both columns (for example, scale and traceability), you would be trading one advantage for another.
- Limit: this activity only uses the six rows of Figure 4. Real projects involve many more factors.

Quick check: Section 5.4 lists a workforce risk of AI automation. What is it, and what example does the section give?

Resources

- **NIST, "AI Risk Management Framework"**: https://www.nist.gov/itl/ai-risk-management-framework. Optional further reading; it is not used as a source in this guide.
- Udacity , "Black Box Testing Example - Georgia Tech - Software Development Process,"

Sources (Chapter 5)

- Chrystal R. China (2024). https://www.ibm.com/think/insights/artificial-intelligence-advantages-disadvantages

Further reading for Chapters 3–5:

- OECD, "Artificial intelligence in society." https://www.oecd.org/en/publications/artificial-intelligence-in-society_eedfee77-en/full-report/component-6.html

- IBM, “What is artificial general intelligence (AGI)?” https://www.ibm.com/think/topics/artificial-general-intelligence
- NIST, “AI Risk Management Framework.” https://www.nist.gov/itl/ai-risk-management-framework

Where Chapters 6–8 fit

Chapters 1–5 told you *what AI is*. The next three chapters answer a deeper question: **how do you actually build something that behaves intelligently?** There are three big answers, and each one makes a different bet.

Approach	The bet	One-line summary
Symbolic AI (Ch. 6)	Intelligence is **reasoning with rules**	Humans write the knowledge down; the machine reasons with it
Connectionist AI (Ch. 7)	Intelligence is **patterns learned from data**	A network of simple units adjusts itself from examples
Behaviourist AI (Ch. 8)	Intelligence is **acting well in the world**	Simple behaviours, or rewards, shape what the agent does

Figure 5. Three approaches to building AI

Prerequisites: Chapters 1–3 (what AI is, the 5-step process, types of AI). No coding experience needed.

Suggested route: read the chapter intro, do the activity *before* reading the explanation if you can, then check your result against the Feedback box.

6. Symbolic AI: Reasoning with Rules

Mission 6: The Rulebook Detective

Challenge: A clinic wants a program that flags possible pneumonia from a patient's symptoms. No training data is available. Could you build it by writing down what doctors know?

6.1 What is Symbolic AI?

Symbolic AI represents knowledge using **symbols**: names for objects and concepts, plus relationships between them. It then reasons by applying logical **rules** to those symbols (DataCamp, 2023). Instead of learning from huge piles of data, a human *writes the knowledge down*.

A classic example is a medical rule of the form:

> IF the patient has a fever AND a cough AND difficulty breathing, THEN the patient may have pneumonia.

The system checks the facts about a patient against its rules and draws a conclusion. A collection of facts and rules like this is called a **knowledge base**, and a program built on one that gives advice in a specialist area is an **expert system**.

6.2 Strengths and limits

Strength	Why
Explainable	You can trace a conclusion back to the exact rules used
Easy to edit	Change a rule and the behaviour changes
Needs little data	The knowledge is written by people, not learned from examples

Limit	Why
Needs well-defined knowledge	If the rules are incomplete, the system can fail
Hard to scale	More symbols and rules can make reasoning very expensive
Poor with uncertainty	It prefers precise, unambiguous facts
Limited learning	It usually must be reprogrammed by hand

Common misconception: "Symbolic AI is old, so it's useless." Not true. Rule-based reasoning is still used where explanations matter, and symbolic and learning-based methods can be **combined** into hybrid systems (DataCamp, 2023).

Link to Chapter 4.3: the expert systems you met earlier are a form of symbolic AI.

Activity 6A: Build a Family Detective (SWISH / Prolog)

Goal	See how conclusions follow from facts and rules, and practise stating relationships precisely.
Preparation	A browser. Open **SWISH** (SWI-Prolog online) at swish.swi-prolog.org and start a new program. SWISH's documentation says it does not require a login by default and only runs safe commands, so no account should be needed. Open it once on your own network before the activity, in case it is blocked.

Task	See below.
Observation	Write down each query, your *prediction*, and the actual answer.
Reflection	Which results surprised you? What does that tell you about the rules you wrote?
Feedback	Compare with the answer box.

Syntax in 30 seconds: `parent(alice, bob).` is a **fact** ("alice is a parent of bob"). Names starting with a lowercase letter are specific things. Uppercase names like `X` are **variables** that Prolog fills in. `:-` means "is true if", and a comma means "and". Every statement ends with a full stop.

Step 1. Type this knowledge base into the program window:

```
parent(tom, alice).
parent(alice, bob).
parent(alice, carol).
parent(bob, dave).

grandparent(X, Z) :- parent(X, Y), parent(Y, Z).
sibling(X, Y) :- parent(P, X), parent(P, Y), X \= Y.
```

Step 2. Predict, then query. Before running each query in the query box, write your prediction.

1. `?- grandparent(tom, X).`
2. `?- grandparent(alice, X).`
3. `?- sibling(bob, X).`
4. `?- sibling(dave, X).`

Step 3. Extend it. Add this rule and predict `?- auntOrUncle(carol, X).` before running it:

```
auntOrUncle(X, Y) :- sibling(X, P), parent(P, Y).
```

Step 4. Break it on purpose. Delete `, X \= Y` from the `sibling` rule and run `?- sibling(bob, X).` again. What changed, and why?

Feedback box

- Query 1 gives `X = bob` and `X = carol` (tom's child alice has children bob and carol).
- Query 2 gives `X = dave`.
- Query 3 gives `X = carol`.
- Query 4 gives `false`: dave's parent bob has no other children.
- `auntOrUncle(carol, X)` gives `X = dave`: carol's sibling is bob, and bob's child is dave.
- Without `X \= Y`, bob counts as his own sibling, because both "bob" slots can match the same child of alice. The rule was *logically valid but incomplete*: it did not say what we obviously assumed. Symbolic AI only knows what you tell it.

Stretch (computational thinking): write facts for a toy symptom checker (`symptom(patient1, fever).` and so on) and a rule `may_have_pneumonia(P)` that needs three symptoms. This is a teaching toy, not medical advice.

Quick check: Why can you *explain* any answer this program gives, but not always explain a neural network's answer? (Hint: look back at Section 5.1.)

Resources

- **SWISH** (SWI-Prolog online): https://swish.swi-prolog.org/. Use it for Activity 6A. It is a teaching environment; real expert systems are much larger.
- **Symbolic AI: Crash Course AI #10** (Jabril Ashe, CrashCourse with PBS Digital Studios, 2019, about 13 minutes): https://www.pbs.org/video/symbolic-ai-10-tfkalm. A transcript summary of the YouTube version puts propositional logic at about 04:46 and the point that expert systems reason with logic rather than trial and error at about 11:36. Viewing question: what is a knowledge base in the video, and how is it like the facts and rules you typed in Activity 6A? (Timestamps are from the YouTube version; the PBS copy is slightly shorter.)
- CrashCourse, Symbolic AI: Crash Course AI #10

Sources (Chapter 6)

- DataCamp, "What is Symbolic AI?" (2023). https://www.datacamp.com/blog/what-is-symbolic-ai
- SWISH documentation (login and sandbox behaviour). https://swish.swi-prolog.org/pldoc/doc/swish/index.html

7. Connectionist AI: Learning from Data

Mission 7: The Canteen Neuron

Challenge: Nobody can write rules for “what does a cat look like in a photo?” What if a program could *learn* the rules by adjusting itself from examples?

7.1 What is connectionism?

Connectionism explains intelligent behaviour using **artificial neural networks** (Stanford Encyclopedia of Philosophy). A network is made of many simple **units** (loosely inspired by brain cells) joined by connections. Each connection has a **weight** that says how strongly one unit influences another. Weights can be positive or negative.

Units are usually grouped into three kinds:

- **Input units** receive the information to be processed.
- **Hidden units** sit in between and pass signals along.
- **Output units** hold the result.

Each unit works out its own activation from the signals it receives, then passes it on. **The knowledge of the network lives in its weights.**

7.2 How does it learn?

Finding good weights is the central goal. Learning methods such as **backpropagation** calculate adjustments to the weights so the network does better on its training examples. Training often takes a huge number of rounds of adjustment (Stanford Encyclopedia of Philosophy).

Misconception check: artificial neural networks are *inspired by* the brain but are **much simpler** than real brains. “Neuron” here means a small calculation, not a living cell.

7.3 Symbolic vs connectionist

	Symbolic	**Connectionist**
Knowledge comes from	Humans writing rules	Patterns learned from data
Explains its answers?	Usually yes	Often hard (the “black box”, Section 5.1)
Needs lots of data?	No	Usually yes
Good at	Logic, structured knowledge	Perception, such as images

7.4 Types of neural networks (your mind map)

Your mind map lists feedforward, convolutional (CNN), recurrent (RNN) and Transformer networks. One beginner-friendly sentence on each:

- **Feedforward:** information moves in one direction, from the input units through the hidden units to the output units, so each layer’s output becomes the next layer’s input (IBM, “What is a neural network?”).

- **Convolutional (CNN):** similar to a feedforward network, but usually used for image recognition, pattern recognition and computer vision (IBM, “What is a neural network?”). The CNN Explainer (Wang et al., 2020) is an interactive tool designed for beginners.
- **Recurrent (RNN):** a deep neural network trained on sequential or time series data, so it can make predictions from sequences (IBM, “What is a recurrent neural network?”).
- **Transformer:** a neural network architecture for sequential data, best known from large language models, built around a self-attention mechanism that picks out relationships between parts of an input (IBM, “What is a transformer model?”).

Activity 7A: Be the Neuron (paper activity)

Goal	Understand that a unit combines weighted inputs, and that *learning means changing weights*.
Preparation	Paper and a pen.
Task	See below.
Observation	Fill in the output for each case, before and after the weight change.
Reflection	Which weight change made the unit more or less willing to “go”?
Feedback	Answer box.

Setup. A neuron decides “Go to the canteen?” (1 = go, 0 = stay). It adds up weighted inputs and goes if the **total is 2 or more**.

Input	Meaning	Weight
Friend going (1 or 0)	A friend is going	**+2**
Raining (1 or 0)	It is raining	**-3**
Hungry (1 or 0)	You are hungry	**+2**

Part 1. Work out the total and decision for each case.

Case	Friend	Rain	Hungry	Total	Go?
A	1	0	0	?	?
B	1	1	0	?	?
C	0	1	1	?	?
D	0	0	1	?	?
E	1	1	1	?	?

Part 2. “Train” the neuron: change the Rain weight from -3 to **-1**. Recalculate cases B, C and E.

Feedback box

- Part 1: A = 2, go. B = -1, stay. C = -1, stay. D = 2, go. E = 1, stay.
- Part 2 (rain weight -1): B = 1, stay. C = 1, stay. E = 3, **go**. Only case E changed.

- Lesson: no rule was rewritten. Only a number changed, and behaviour changed with it. Real networks do the same with huge numbers of weights, adjusted automatically by learning algorithms.
- Limit of this analogy: a real network learns its weights from many examples; you set them by hand here.

Activity 7B: Playground Lab (TensorFlow Playground)

Goal	Predict and observe how network settings change learning.
Preparation	Open TensorFlow Playground (playground.tensorflow.org) in a browser. Note the date you used it, since the controls may change.
Task	Pick one dataset. Predict whether a network with **no hidden layer** can separate it. Run it. Then add one hidden layer and compare. Change **one setting at a time** (for example learning rate, or number of neurons).
Observation	For each run, record the setting you changed, the training loss and the test loss, and whether the shape of the boundary changed.
Reflection	Which change helped most? Did any change make test loss worse than training loss?
Feedback	Expected outcomes below.

Feedback box

- With no hidden layer and only the raw x and y inputs, the network can only draw a straight-line boundary. Expect it to separate data that a straight line can split, and to struggle with the circle, exclusive-or and spiral datasets.
- Adding a hidden layer with a few neurons lets the boundary bend, so the circle and exclusive-or datasets usually become separable. The spiral is the hardest and usually needs more neurons or more layers.
- A very high learning rate can make the loss jump around instead of settling; a very low one learns slowly. More neurons help up to a point.
- If the test loss ends up much higher than the training loss, the network may be fitting the training points too closely.
- These are the usual behaviours of this tool, written from general knowledge of how it works rather than from your exact settings, so treat them as predictions to test against your own runs.

Fallback if the site is unavailable: include 2–3 screenshots of runs and ask learners to explain the difference between them.

Resources

- **CNN Explainer**, an interactive visualization of convolutional networks (Wang et al., 2020): https://poloclub.github.io/cnn-explainer/. Use it after Activity 7B if you cover CNNs. It assumes a basic idea of images as grids of numbers.
- **Neural Networks and Deep Learning: Crash Course AI #3** (Jabril Ashe, CrashCourse with PBS Digital Studios, 2019): https://thecrashcourse.com/courses/neural-networks-and-deep-learning-crash-course-ai-3/. The episode combines the single artificial neuron into a neural network and explains how many hidden layers allow deep learning. Viewing question: what do hidden layers add compared with the single neuron in Activity 7A? Watch the whole episode.

Sources (Chapter 7)

- Stanford Encyclopedia of Philosophy, “Connectionism”. https://plato.stanford.edu/entries/connectionism/
- Wang, Z. J., Turko, R., Shaikh, O., Park, H., Das, N., Hohman, F., Kahng, M. and Chau, D. H. (2020), “CNN Explainer: Learning Convolutional Neural Networks with Interactive Visualization”, arXiv:2004.15004. https://arxiv.org/pdf/2004.15004
- IBM, “What is a neural network?”. https://www.ibm.com/topics/neural-networks
- IBM, “What is a recurrent neural network (RNN)?”. https://www.ibm.com/topics/recurrent-neural-networks
- IBM, “What is a transformer model?”. https://ibm.com/think/topics/transformer-model

8. Behaviourist AI: Acting and Learning from Rewards

Mission 8: The Confused Vacuum

Challenge: A robot vacuum must avoid furniture, return to its dock, and still clean the floor. Do you need a giant world model, or can simple behaviours stacked together do the job?

> **Two ideas, one chapter.** (1) **Behaviour-based AI** builds intelligent action from layered simple behaviours. (2) **Reinforcement learning** learns good actions from rewards. They are different techniques; both focus on *behaviour* rather than on written rules or stored knowledge.

8.1 Behaviour-based AI

By the 1980s, researchers found it hard to use symbolic AI in real robots. This led to a "reactive" approach: low-level behaviours that respond to the world directly, without building a detailed internal model first. Rodney Brooks introduced the **subsumption architecture** for this (Moore, 2016).

- A subsumption architecture is a **hierarchy of simple behaviours** (Wooldridge, Parsons and Payne, course slides).
- Each behaviour is a rather simple, rule-like structure, and complex behaviour **emerges** from simple components interacting with the environment (same slides).
- Higher layers can take over (subsume) lower ones when appropriate (University of York Brooks page).

Brooks later founded iRobot, the company behind the Roomba robot vacuum, but the subsumption approach has been criticised as hard to scale to more complex robots (Moore, 2016).

Misconception check: "Behaviour-based means no rules." Not quite. The behaviours are simple rules; the difference is that there is no large symbolic model of the world.

Activity 8A: Layer the Vacuum (paper activity)

Goal	See how priority layers of simple behaviours produce sensible action.
Preparation	Paper and a pen.
Task	A toy robot has three behaviours, ranked from **highest to lowest priority**: (1) **Avoid**: if the bumper is pressed, turn away. (2) **Dock**: if the battery is low, head to the charger. (3) **Wander**: otherwise, move forward. For each situation, decide which behaviour takes control.
Observation	Fill in the table.
Reflection	What happens if you swap the priority of Avoid and Dock?
Feedback	Answer box.

Situation	Bumper pressed?	Battery low?	Which behaviour wins?
1	No	No	?
2	No	Yes	?
3	Yes	No	?

Situation	Bumper pressed?	Battery low?	Which behaviour wins?
4	Yes	Yes	?

Feedback box

- 1 = Wander. 2 = Dock. 3 = Avoid. 4 = Avoid (highest priority wins).
- Swapping priorities: in situation 4 the robot would drive toward the charger *into* the obstacle. Priority order is a design decision, and wrong priorities cause bad behaviour.
- Limit: this is a **simplified teaching model** inspired by subsumption, not a description of how any real vacuum works.

8.2 Reinforcement learning

In **reinforcement learning (RL)**, an **agent** learns by interacting with an **environment**. It takes an action, the environment responds with a new situation and a **reward** (or penalty), and the agent adjusts to earn more reward over time. The strategy it learns is called a **policy**. Unlike supervised learning, RL does not rely on labelled examples; it learns by **trial and error** (OVHcloud; Verpex).

Two ideas to remember:

- **Exploration vs exploitation:** try new things, or repeat what already worked?
- **Delayed rewards:** some actions pay off only later, which makes learning harder.

Link to the guide: the “Adjustments” step in Chapter 2 is a loose picture of feedback improving behaviour. RL is one concrete way that happens.

Activity 8B: Three Doors (dice game, exploration vs exploitation)

Goal	Feel the exploration/exploitation trade-off.
Preparation	One die per pair, paper, 10 minutes.
Task	You are an agent choosing among 3 doors for 15 rounds. **Rules (hidden from the player; the partner reads them):** Door A always pays +1. Door B pays +5 if the die shows 6, otherwise 0. Door C pays +3 if the die shows 4, 5 or 6, otherwise 0. The player picks a door, the partner rolls (if needed) and announces the reward. Keep a tally.
Observation	After 15 rounds record total reward, how many times you tried each door, and when you decided which door was best.
Reflection	Did you stop exploring too early? What would a better strategy be?
Feedback	Answer box.

Feedback box

- Average reward per round: A = 1; B = 5 x 1/6, about 0.83; C = 3 x 1/2 = 1.5. So **C is best on average**.
- Over only a few rounds luck can mislead you: B might pay +5 on the first try and look best. That is exactly why agents must keep exploring for a while before committing.

- This is a simplified RL-style game (there is only one situation, so no “state”). Full RL adds changing situations and delayed rewards.

Resources

- **Reinforcement Learning: Crash Course AI #9** (Jabril Ashe, CrashCourse with PBS Digital Studios, 2019, about 11 minutes): https://www.pbs.org/video/reinforcement-learning-9-t9yhid. A transcript summary of the YouTube version puts the introduction to learning by trial and error at about 00:25 and the balance between exploration and exploitation at about 08:59. The episode also calls the problem of deciding when to reward an AI “credit assignment”. Viewing question: in Activity 8B, which of your choices were exploration and which were exploitation? (Timestamps are from the YouTube version; the PBS copy is slightly shorter.)

Sources (Chapter 8)

- Wooldridge, M., Parsons, S. and Payne, T. R., “COMP310 Chapter 5” (University of Liverpool lecture slides, 2018). https://www.csc.liv.ac.uk/~trp/COMP310_files/COMP310%20Chapter5%202018.pdf
- Moore, R. K. (2016), “PCT and Beyond: Towards a Computational Framework for ‘Intelligent’ Communicative Systems”, arXiv:1611.05379. https://arxiv.org/pdf/1611.05379
- University of York, page on Rodney A. Brooks (Susan Stepney’s book pages). https://www-users.cs.york.ac.uk/susan/books/pages/b/RodneyABrooks.htm
- OVHcloud, “What is reinforcement learning?”. https://www.ovhcloud.com/en-in/learn/what-is-reinforcement-learning/
- Verpex, “Reinforcement Learning: How Machines Learn Through Rewards” (2025). https://verpex.com/blog/website-tips/what-is-reinforcement-learning

Supplementary Resources

For more practice checkout: https://kahoot.it/challenge/03772119?challenge-id=57467697-25a7-424c-9f82-3967c1284b04_1794033556332

For detailed video explanations check out our own video tutorials:
https://drive.google.com/drive/folders/1lsnS5zVKpcZWO3ieUxUFK0_NMUHiegBL

Glossary

Term	Meaning
Artificial intelligence (AI)	A foundational technology with the potential to transform societies, economies and politics
Narrow AI	AI trained to perform a single or narrow task; the only type that exists today
Artificial general intelligence (AGI)	Theoretical AI that could learn and perform any intellectual task a human can
Super AI	Theoretical AI whose abilities would surpass those of human beings
Computer vision	Narrow AI trained to interpret and classify images and video
Black box	A model whose input and output are known but whose inner process is unclear
Unstructured data	Data such as images, video and natural language text
Symbolic AI	AI that reasons over symbols using human-written rules
Knowledge base	A collection of facts and rules a symbolic system reasons with
Expert system	A rule-based system that gives advice in a specialist area
Connectionism	The approach that builds AI from networks of simple connected units
Weight	A number that sets how strongly one unit influences another
Backpropagation	A learning method that adjusts weights to reduce errors
Subsumption architecture	A hierarchy of simple behaviours where higher layers can override lower ones
Emergent behaviour	Complex behaviour arising from simple parts interacting with the world
Reinforcement learning	Learning actions by trial and error using rewards and penalties
Agent / Environment / Reward / Policy	The learner / what it acts in / its feedback / its learned strategy